API SPECIFICATION

Grid-Guard AI

Backend REST API Contract

Request/response schemas for every endpoint consumed by the web dashboard and mobile app

Field	Value
Author	Noor Eman
Version	1.0 (Draft)
Base URL	https://api.grid-guard.example/v1 (placeholder)
Auth	Bearer JWT in Authorization header, except /auth/login

1. Conventions

- All request/response bodies are JSON unless noted (photo upload uses multipart/form-data).
- All timestamps are ISO 8601 UTC.
- All list endpoints are paginated via ?page= and ?page_size= query params (default page_size=25).
- Standard error shape (all non-2xx responses):

```
{
  "error": {
    "code": "NOT_FOUND",
    "message": "Case not found"
  }
}
```

2. Auth

POST /auth/login

Request:

```
{ "email": "sana@lesco.example", "password": "..." }
```

Response 200:

```
{
  "token": "eyJhbGciOi...",
  "user": { "id": "uuid", "name": "Sana", "role": "officer" }
}
```

Response 401: invalid credentials, standard error shape.

3. Consumers & Risk Queue

GET /consumers/risk-queue

Used by: Web (WEB-1). Query params:

Param	Type	Notes
feeder	string, optional	Filter by feeder name
risk_band	enum, optional	high | medium | low
meter_type	enum, optional	conventional | ami
page / page_size	int, optional	Pagination

Response 200:

```
{
  "results": [
    {
      "consumer_id": "uuid",
      "external_ref": "4471",
      "address": "Street 12, Model Town",
      "feeder": "Model Town-04",
      "risk_score": 0.91,
```

```
      "risk_band": "high",
      "is_demo_data": true
    }
  ],
  "page": 1,
  "page_size": 25,
  "total": 132
}
```

GET /consumers/{id}/explanation

Used by: Web (WEB-2), Mobile (MOB-2). Response 200:

```
{
  "consumer_id": "uuid",
  "risk_score": 0.91,
  "risk_band": "high",
  "explanation_text": "Consumption dropped 60% in the last 3 months compared to seasonal baseline, without change in appliance ownership on record.",
  "trend_series": [ { "period": "2026-05", "usage_kwh": 210 }, { "period": "2026-06", "usage_kwh": 90 } ]
}
```

Response 404: consumer not found, standard error shape.

4. Cases

POST /cases

Used by: Web (WEB-3). Creates and assigns a case in one step. Request:

```
{
  "consumer_id": "uuid",
  "risk_score_id": "uuid",
  "assigned_to": "uuid",
  "assignment_note": "Priority — check today if possible."
}
```

Response 201:

```
{
  "case_id": "uuid",
  "status": "assigned",
  "created_at": "2026-08-25T09:22:00Z"
}
```

GET /cases

Used by: Web (WEB-4). Query params: status (optional filter), feeder (optional), page/page_size.

```
{
  "results": [
    {
      "case_id": "uuid",
      "consumer_ref": "4471",
      "assigned_to_name": "Bilal Ahmed",
      "status": "in_progress",
      "created_at": "2026-08-25T09:22:00Z"
    }
  ],
  "page": 1, "page_size": 25, "total": 4
}
```

GET /cases/mine

Used by: Mobile (MOB-1). Returns cases where assigned_to = current authenticated user. Same response shape as GET /cases, no feeder/status filters needed for v1.

GET /cases/{id}

Used by: Web, Mobile (MOB-2). Full case detail including consumer address, meter ID, and the linked explanation (same shape as /consumers/{id}/explanation, embedded).

GET /cases/{id}/audit-log

Used by: Web (WEB-5). Response 200:

```
{
  "entries": [
    { "action": "flag_generated", "actor": "system", "at": "2026-08-25T09:14:00Z"
},
    { "action": "case_assigned", "actor": "Sana", "at": "2026-08-25T09:22:00Z" },
    { "action": "inspection_submitted", "actor": "Bilal Ahmed", "at": "2026-08-
25T17:40:00Z" }
  ]
}
```

5. Inspections

POST /cases/{id}/inspection

Used by: Mobile (MOB-3, MOB-4). Content-Type: multipart/form-data (photo is a binary file part; other fields are form fields).

Field	Type	Notes
outcome	enum	confirmed | false_positive | inconclusive
tampering_type	string[]	Sent as repeated form fields or comma-separated
note	string	Optional
gps_lat / gps_lng	float	Captured on-device at submission time
photo	file	JPEG, captured on-device
submitted_at	ISO 8601 string	Device-local submission time — may predate the server's receipt time if submitted offline

Response 201:

```
{
  "inspection_id": "uuid",
  "case_id": "uuid",
  "case_status": "closed_confirmed",
  "synced_at": "2026-08-25T18:02:00Z"
}
```

This endpoint is idempotent on repeated identical submission (same case_id + submitted_at) to safely tolerate the mobile app's retry-on-reconnect sync logic without creating duplicate inspection records.

6. Model Performance

GET /model-performance

Used by: Web (WEB-7). Response 200:

```
{
  "confirmed_rate": 0.63,
  "false_positive_rate": 0.21,
  "inconclusive_rate": 0.16,
  "trend": [
    { "month": "2026-06", "confirmed_rate": 0.58 },
    { "month": "2026-07", "confirmed_rate": 0.63 }
  ]
}
```

Calculated only over cases with status starting with closed_ — open/in-progress cases are excluded from the rate calculation.

7. HTTP Status Code Summary

Code	Meaning in this API
200	Successful GET
201	Resource created (case, inspection)
400	Malformed request body / missing required field
401	Missing or invalid JWT
403	Valid JWT but insufficient role (e.g. inspector calling an officer-only endpoint)
404	Resource not found
409	Conflict — e.g. attempting to assign an already-closed case
500	Unhandled server error